Obstructions and a Corrected Spectral Criterion for GRH

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Research draft. GRH is not proved here.

Abstract

We audit proposed surface-order and Hilbert–Pólya arguments for GRH. We retain a conditional spectral-approximation criterion and prove three obstructions: integral hyperbolic matrices have no integer prime-power norms, integral cyclic induction fails already for the Klein four group, and absolute prime tails diverge throughout the proposed critical-strip range. We also exhibit determinant information lost by symmetrization and a direct failure of the supplied diagonal diagnostic. We supply convergent arithmetic replacements using Dirichlet partial sums and pole-subtracted zeta sums, with explicit errors. These results identify what a complete arithmetic spectral construction must still establish. They do not establish GRH. A Lean 4 companion verifies exact arithmetic obstructions, complex Euler-factor nonvanishing, and a local boundary-convergence lemma. The full arithmetic spectral bridge remains unresolved.

1 Scope and conventions

For a primitive Dirichlet character χ of conductor q, write

$$L(s,\chi)=\sum_{n\ge 1}\frac{\chi(n)}{n^s},\qquad \Lambda(s,\chi)=\left(\frac{q}{\pi}\right)^{(s+a)/2}\Gamma\left(\frac{s+a}{2}\right)L(s,\chi),\quad a=\frac{1-\chi(-1)}{2}.$$

The series and Euler product initially apply for $\operatorname{Re} s>1$. Our target is Dirichlet GRH: every zero in $0<\operatorname{Re} s<1$ has real part $1/2$. For $q=1$, use the entire function $\xi(s)=\frac{1}{2}s(s-1)\Lambda(s,1)$ when an entire target is required. Finite-order Hecke characters and general Artin representations require separate constructions; a theorem for one family is not silently a theorem for another.

The supplied surface-order manuscripts [1, 2] propose arithmetic transfer operators and self-adjoint approximants. We isolate valid closure logic and prove obstructions to particular claimed constructions. These obstructions refute intermediate statements, not GRH. The separate `math` repository [3] includes claimed zero-free regions strictly weaker than GRH. Such a region cannot replace the missing critical-line argument.

2 A corrected conditional closure theorem

Theorem 2.1 (Spectral approximation criterion)**.** *Let $U\subset\mathbb{C}$ be connected and open, let f be holomorphic on U and not identically zero, and let*

$$F_N(s)=e^{a_N+b_N s}\det\bigl(sI-(\tfrac{1}{2}I+iH_N)\bigr),\qquad H_N=H_N^*\in M_{d_N}(\mathbb{C}).$$

If $F_N\to f$ locally uniformly on U, then every zero of f in U has real part $1/2$.

Proof. The eigenvalues $\lambda_{j,N}$ of H_N are real. Thus the zeros of F_N are $1/2 + i\lambda_{j,N}$, with algebraic multiplicities; the exponential factor never vanishes. If $f(z) = 0$ off that line, choose a closed disk $D \subset U$ about z disjoint from the line, with f nonzero on its boundary. Such a disk exists because the zeros of a nonzero holomorphic function on a connected domain are isolated. Let $m = \min_{\partial D} |f| > 0$. Local uniform convergence gives $|F_N - f| < m$ on ∂D for sufficiently large N. Rouché's theorem forces F_N and f to have the same positive number of zeros inside D, contradicting the zero location of F_N. □

Remark 2.2. This is a conditional criterion, not a construction. Self-adjointness alone establishes only the zero location of the approximants. The difficult arithmetic statement is convergence to the particular completed L-function. Pointwise convergence, weak operator convergence, and agreement of finitely many zeros do not supply the displayed hypothesis.

Proposition 2.3 (A usable continuation interface)**.** *Suppose U is a connected domain, the entire functions F_N are locally uniformly bounded on U, and they converge pointwise on a nonempty open subset $V \subset U$ to a holomorphic function f defined on U. Then $F_N \to f$ locally uniformly on U.*

Proof. Montel's theorem makes the family normal. Every subsequence has a further subsequence converging locally uniformly to a holomorphic function g. On V, pointwise convergence gives $g = f$, so the identity theorem gives $g = f$ on U. If the original sequence failed to converge locally uniformly, a compact set, positive error, and violating subsequence would contradict this subsequence property. □

A proposed analytic continuation proof must actually provide the uniform bounds and a genuine overlap V. The disjoint domains $\operatorname{Re} s > 1$ and $0 < \operatorname{Re} s < 1$ are not an overlap.

3 Arithmetic obstructions to the claimed completion

Theorem 3.1 (No integral prime-power geodesic norms)**.** *Let $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ be hyperbolic. Its expanding eigenvalue in absolute value is $u > 1$, and its hyperbolic norm is $N(\gamma) = u^2$. Then $N(\gamma)$ is not an integer greater than one. In particular no closed geodesic arising from such a matrix has length $k \log p$ for a prime p and integer $k \geq 1$.*

Proof. Put $t = |\operatorname{tr} \gamma| \in \mathbb{Z}$, so $t = u + u^{-1}$. If $u^2 = n \in \mathbb{Z}$, $n > 1$, squaring gives

$$t^2 = n + 2 + \frac{1}{n}.$$

The left-hand side and $n + 2$ are integers, so $1/n$ is an integer, contradicting $0 < 1/n < 1$. Since length is $\log N(\gamma)$, length $k \log p$ would imply $N(\gamma) = p^k$, already excluded. □

The statement applies to every subgroup of $\mathrm{SL}_2(\mathbb{Z})$, hence to congruence surfaces. The prime geodesic theorem is an asymptotic counting analogy; it cannot provide the termwise identification claimed in the completion section of [1]. Hecke correspondences of determinant p must not be identified with hyperbolic deck transformations of determinant one.

Theorem 3.2 (Failure of integral cyclic induction)**.** *The trivial character of $G = C_2 \times C_2$ is not an integer linear combination of characters induced from one-dimensional characters of cyclic subgroups of G.*

Proof. Every cyclic subgroup has order one or two. The degree of any such induced character is its index, namely four or two. Evaluation at the identity sends every integer linear combination to an even integer, whereas the trivial character has degree one. □

This distinguishes rational Artin induction from integral Brauer induction. It invalidates the stated cyclic integral reduction, while leaving appropriate forms of these classical induction theorems available for a revised argument.

Theorem 3.3 (Prime-tail obstruction)**.** *For every real $\sigma \le 1$, the positive series $\sum_p p^{-\sigma}$ diverges. Thus $\sum_{p>N} p^{-\sigma}$ is not a vanishing absolutely convergent tail in the critical strip.*

Proof. It suffices to show $\sum_p 1/p = \infty$, since $p^{-\sigma} \ge p^{-1}$. Suppose the reciprocal-prime series converged. For $0 \le x \le 1/2$, $-\log(1-x) \le 2x$, for example by integrating $(1-t)^{-1} \le 2$ from zero to x. Hence the finite Euler products $\prod_{p \le X}(1-1/p)^{-1}$ would be bounded uniformly in X. Expanding each as a finite product of absolutely convergent geometric series gives the sum of $1/n$ over all positive integers with prime factors at most X. In particular it dominates $\sum_{n \le \lfloor X \rfloor} 1/n$, which is unbounded. Contradiction. □

Any repair below $\operatorname{Re} s = 1$ must introduce proved cancellation or a regularization, with quantitative control; absolute prime-tail bounds cannot suffice.

4 Determinants and lost arithmetic information

For $T = \operatorname{diag}(\chi(p)p^{-s})_{p \le N}$ one has exactly

$$\det(I - T) = \prod_{p \le N} (1 - \chi(p)p^{-s}) = L_N(s, \chi)^{-1},$$

not L_N. On an absolute-convergence region,

$$\log\det(I - T) = -\sum_{k \ge 1} \frac{\operatorname{tr}(T^k)}{k}.$$

The reciprocal orientation matters before any analytic continuation. Moreover Hermitian-part extraction does not preserve this determinant:

$$T = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \frac{T + T^*}{2} = 0, \quad \det(I - T) = 2 \ne 1 = \det(I).$$

An identity for the transfer determinant therefore does not automatically identify the centered characteristic polynomial of a symmetrized operator.

Proposition 4.1 (A direct obstruction to the supplied diagonal diagnostic)**.** *For the $q = 1$ model evaluated at $s = 1/2$, the algorithm that sets $\lambda_p = -\arg(p^{-1/2})$ gives $H_N = 0$ and $F_N(s) = (s - 1/2)^{d_N}$. These functions cannot converge locally uniformly to $\Lambda(s, 1)$ or $\xi(s)$ on a neighborhood of $1/2$.*

Proof. Every $p^{-1/2}$ is positive, so its argument is zero and $F_N(1/2) = 0$. On the other hand the alternating Dirichlet series $\eta(1/2) = \sum_{n \ge 1}(-1)^{n-1}n^{-1/2}$ is positive: grouping into adjacent pairs gives positive terms and its sum is at least $1 - 1/\sqrt{2} > 0$. The continuation identity $\eta(s) = (1 - 2^{1-s})\zeta(s)$ gives $\zeta(1/2) < 0$. The gamma factor at $1/4$ is positive and finite, so $\Lambda(1/2, 1) \ne 0$, and its pole-removal factor at $1/2$ is nonzero. Even pointwise convergence at that point is impossible. □

5 A necessary change of approximation class

Theorem 5.1 (Zero-free Euler truncations cannot approximate a zero)**.** *Let U be a connected open subset of $\{\operatorname{Re} s > 0\}$, let g_N be holomorphic and nowhere zero on U, and let P_N be arbitrary finite sets of rational primes. Put*

$$D_N(s) = \prod_{p \in P_N} (1 - \chi(p) e^{-s \log p}).$$

Each of $g_N D_N$ and g_N / D_N is holomorphic and nowhere zero on U. If either sequence converges locally uniformly to a holomorphic f which is not identically zero, then f has no zero in U.

Proof. For $\operatorname{Re} s > 0$, $|\chi(p) e^{-s \log p}| \le p^{-\operatorname{Re} s} < 1$, so every local factor is nonzero. Hence the finite product, its reciprocal, and their normalized versions have the asserted property. If f had a zero, choose a closed disk inside U with a positive number of zeros of f in its interior and none on its boundary. Local uniform convergence and Rouché's theorem would force the zero-free approximants to have that positive number of zeros, a contradiction. □

More quantitatively, on any such disk D, every zero-free holomorphic approximant G satisfies

$$\max_{\partial D} |G - f| \;\ge\; \min_{\partial D} |f| > 0.$$

Otherwise Rouché yields the same contradiction. Thus improved estimates cannot repair this approximation class near a nontrivial zero. The construction must change so that finite approximants can have zeros in the strip; a nonvanishing gamma or exponential normalization cannot do this.

6 A convergent arithmetic replacement

For a nonprincipal character modulo q, put $A(x) = \sum_{n \le x} \chi(n)$ and $B = \sup_x |A(x)| \le q$. Complete periods sum to zero, so B is finite. Partial summation gives, for $\sigma = \operatorname{Re} s > 0$,

$$L(s, \chi) - \sum_{n \le N} \chi(n) n^{-s} = -A(N) N^{-s} + s \int_N^\infty A(x) x^{-s-1} \, dx,$$

and therefore

$$\left| L(s, \chi) - \sum_{n \le N} \chi(n) n^{-s} \right| \le B \left(1 + \frac{|s|}{\sigma} \right) N^{-\sigma}.$$

The endpoint tends to zero and the integral is absolutely convergent. The limit agrees with the Euler definition on $\operatorname{Re} s > 1$, hence with its analytic continuation. Multiplication by the correct gamma/conductor factor gives locally uniform approximation on $\operatorname{Re} s > 0$ for either parity.

For zeta, reproduce the pole before removing it. Set

$$Z_N(s) = \sum_{n \le N} n^{-s} + \frac{N^{1-s}}{s-1}.$$

Euler summation gives, for $\sigma > 0$ and $s \ne 1$,

$$\zeta(s) - Z_N(s) = -s \int_N^\infty \{x\} x^{-s-1} \, dx, \qquad |\zeta(s) - Z_N(s)| \le \frac{|s|}{\sigma} N^{-\sigma}.$$

The integral continues holomorphically to $\sigma > 0$. Consequently

$$\xi_N(s) = \frac{s}{2}\pi^{-s/2}\Gamma(s/2)\left((s-1)\sum_{n\le N} n^{-s} + N^{1-s}\right)$$

is holomorphic on that half-plane, including $s = 1$, and converges locally uniformly to ξ; in fact $\xi_N(1) = 1/2$ for every N. These constructions use known arithmetic coefficients, not unknown zeros. They avoid Theorem 5.1 by allowing additive cancellation and unrestricted finite-level zeros. They establish convergence, but provide no self-adjoint realization or critical-line theorem for those zeros.

7 Consequences for a revised research article

Theorems 3.1, 3.2, and 3.3 require replacements for the associated proof steps, not additional exposition. A credible next theorem would construct operators independently of unknown zeros and establish their exact arithmetic determinant, including ramified, archimedean, and conductor terms. It must then prove locally uniform convergence or the local boundedness in Proposition 2.3. Only then does Theorem 2.1 prove the desired zero location.

Weil/Li positivity, adelic trace methods, and canonical-product approximations are potential alternative interfaces. They move the research obligation to a specific positivity, trace, or approximation theorem; they do not discharge it by analogy. Lagarias [4] and Connes [5] provide relevant primary-source starting points. No novelty or complete proof is claimed for these classical conditional principles.

References

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[4] J. C. Lagarias, Li coefficients for automorphic L-functions, *Ann. Inst. Fourier* **57** (2007), 1689–1740, `https://arxiv.org/abs/math/0404394v4`.

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[6] The mathlib community, *Riemann zeta function documentation*, `https://leanprover-community.github.io/mathlib4_docs/Mathlib/NumberTheory/LSeries/RiemannZeta.html`.